Pure cuts and the stable counting correspondence

Human Compression project

All groups in the proof are actual permutation groups. The notation $\mathrm{Sym}(X)$ means the full symmetric group supported on X, fixing its complement pointwise. A perfect cut of a transitive string C-marking has two omission orbits A, B, with the preceding and following label intervals supported and transitive on A, B, respectively. Its mark is $(a\ b)\alpha\beta$, where the internal factors are supported on $A \setminus \{a\}$ and $B \setminus \{b\}$. A pure cut has mark $(a\ b)$. Internal cuts exclude the two endpoint labels.

We acquire the exact leaf-kernel criterion, the actual-wing bounds and the complete sufficient-split theorem. The latter gives, when $2r > n + 2$, an internal pure cut or an internal perfect cut with one trivial internal factor and two odd actual attaching groups. It does not assert that every usable cut admits subdivision. We shall use pure cuts throughout the counting argument.

Lemma 1 (A symmetric factor admitting a pure leaf)**.** *Let $H = \langle h_0, \dots, h_{k-1} \rangle = \mathrm{Sym}(W)$ be a natural symmetric string C-marking, with a moved by h_0 and fixed by the later marks. If adjoining $(a\ c)$ at a new point gives a C-marking, then h_0 is a pure transposition. More generally the same conclusion follows from the full leaf-kernel tests at a.*

Proof. Put $P_j = \langle h_0, \dots, h_{j-1} \rangle$ and $X_j = a^{P_j}$, with $P_0 = 1$. The acquired exact criterion gives

$$P_j \cap \mathrm{Sym}(X_{j-1}) \le P_{j-1} \qquad (1 \le j \le k).$$

We have $P_k = \mathrm{Sym}(W) \supseteq \mathrm{Sym}(X_k)$. Descending on j, if P_j contains $\mathrm{Sym}(X_j)$, it contains $\mathrm{Sym}(X_{j-1})$, and the jth test puts the latter group in P_{j-1}. Hence every P_j contains $\mathrm{Sym}(X_j)$. In particular $\langle h_0 \rangle$ contains the pure transposition on $X_1 = \{a, a^{h_0}\}$. Since it has order two, that transposition is h_0 itself. □

Theorem 2 (Existence of an internal pure cut)**.** *A fractured natural symmetric string C-marking of degree $n \ge 5$ and rank r with $2r > n+2$ has an internal pure perfect cut. For $n \ge 7$ fracture is automatic. In degrees five and six the ordinary simplex representative gives the same conclusion on marked-group isomorphism classes.*

Proof. Apply the acquired sufficient-split theorem. If its cut is not pure, reverse the order if necessary and write it as $q = (a\ b)\beta$, with supported transitive wings L, R on A, B. The attaching groups L and $J = \langle q, R \rangle$ are both odd. Set $p = \mathrm{rank}(L)$ and $s = \mathrm{rank}(R)$. Then

$$(2p - |A|) + (2s - |B|) > 0. \tag{1}$$

If $2s > |B|$, the actual-wing theorem gives $R = \mathrm{Sym}(B)$. Since $\beta \in R$, it follows that $J = \mathrm{Sym}(\{a\} \cup B)$. The original IP and the odd external group L give the full leaf-kernel tests for J at a, by the acquired odd-attachment lemma. Lemma 1 makes its first mark q pure, contrary to this case.

Thus $2p > |A|$, and $L = \mathrm{Sym}(A)$. Apply the same full-kernel argument to L, oriented from its last mark, with odd external J. Its last mark is pure, say $(d\ a)$. All earlier L marks fix a. Their graph on $A \setminus \{a\}$ is connected: deleting a pendant vertex from the connected graph of L preserves connectedness. They are therefore transitive on that set. In the whole marking the opposite interval is J, supported and transitive on $\{a\} \cup B$. Hence $(d\ a)$ is a pure perfect cut of

the whole marking. It is internal: $2p > |A|$ forces $p \geq 2$, since a single transitive involution has degree two; the original q and its nonempty suffix follow it.

The acquired initial-fracture theorem supplies fracture for $n \geq 7$. At degrees five and six the strict rank inequality forces the simplex up to marked-group isomorphism. The outer degree-six image is replaced by the ordinary action, exactly as in that acquired classification. □

Lemma 3 (The interval of pure cuts)**.** *In any transitive string C-marking, every label between two pure perfect cuts is itself a pure perfect cut. Consecutive such cuts form a simple path, whose internal vertices are fixed by every mark outside that path.*

Proof. Let the cuts have labels $i < j$, with omission supports A_i, B_i and A_j, B_j. We have $A_i \subset A_j$ and $B_j \subset B_i$: the earlier prefix is contained in the later prefix, and the pure i-edge, also in the latter, joins its support to one further point. This also proves the inclusion when the earlier prefix is trivial on a singleton. The reversed argument gives the other inclusion. Put $W = A_j \setminus A_i = B_i \setminus B_j$. Write the two marks as $u = (a\ b)$ and $v = (c\ d)$, with $a \in A_i$, $b, c \in W$ and $d \in B_j$.

If $j = i + 1$, the later prefix is generated by the earlier one and u. Its orbit is exactly $A_i \cup \{b\}$, so $W = \{b\}$ and $b = c$. Every other mark fixes this point, by the two separate support conditions.

Suppose $j > i + 1$, and let M be the intervening label subgroup. It is supported on W. It is transitive there: the suffix at i is generated by M, v and the suffix at j, and its sole edge from W to the latter's orbit B_j is cd. Thus every point of W lies in the M-orbit of c. The pure-leaf factorization now gives the actual supported groups

$$\langle u, M\rangle = \mathrm{Sym}(\{a\} \cup W), \qquad \langle M, v\rangle = \mathrm{Sym}(W \cup \{d\}).$$

Their original label sets intersect exactly in those of M. The original IP therefore gives $M = \mathrm{Sym}(W)$.

The first mark of M moves b, and all its later marks fix b by commutation with u. Lemma 1, applied to the old C-submarking (u, M), makes that first mark pure. It is a perfect cut globally: its prefix is the old prefix joined through u to b, and its suffix is the intervening tail on $W \setminus \{b\}$ joined through v to B_j. The tail is transitive after deleting the pendant b. Here $b \neq c$, since the distant pure marks u, v commute. The argument also covers a one-mark intervening subgroup, whose remaining support is a singleton.

Repeat with this new cut and j. This proves the interval assertion. The adjacent-cut calculation shows that successive middle vertices are fixed by every other mark. They are distinct: each new prefix orbit gains exactly its new middle vertex. Thus the cuts are a simple path with the stated support property. □

Lemma 4 (The first internal pair)**.** *If $2r > n + 2$ and an internal pure perfect cut exists, the least internal pure index i is followed by another internal pure perfect cut at $i + 1$.*

Proof. Let its wing degrees be a, b and ranks $p = i$, $s = r - i - 1$. Then $(2p - a) + (2s - b) > 0$. If $2p > a$, the actual-wing theorem makes the prefix $\mathrm{Sym}(A)$. Its old pure-leaf extension by the i-mark is C. Lemma 1 and the pendant argument in Theorem 2 give a pure perfect cut at $i - 1$. It is internal because $p \geq 2$, contradicting the choice of i. Hence $2s > b$. The same argument on the symmetric suffix gives a pure perfect cut at $i + 1$. Since $s \geq 2$, this cut is internal. □

Lemma 5 (Contraction of a pure pair)**.** *Two consecutive pure perfect cuts may be contracted to one pure perfect cut. The result is a string C-marking of the full symmetric group on one fewer point, and has one fewer mark.*

Proof. The adjacent geometry in Lemma 3 gives disjoint sets $A, \{c\}, B$, supported transitive preceding and following groups L, T, and the two pure marks $(a\ c), (c\ b)$. Every other mark fixes

c. Replace this pair by $q = (a\ b)$ and delete c. The new mark is distinct from all old ones, which are supported separately on A, B. Every old distant mark fixes the corresponding endpoint by the old string relations, so the new string relations hold.

We pay IP through actual intersections. For subgroups $P \le \mathrm{Sym}(A)$ and $Q \le \mathrm{Sym}(B)$ put $X = a^P$, $Y = b^Q$ and $H(P,Q) = \langle P, Q, (a\ b)\rangle$. Conjugation of $(a\ b)$ by P, Q gives every $(x\ y)$ with $x \in X, y \in Y$. These generate the symmetric group on $X \cup Y$. Cancelling restrictions on that set gives the direct-product identity

$$H(P,Q) = \mathrm{Sym}(X \cup Y) \times P^{A \setminus X} \times Q^{B \setminus Y}. \tag{2}$$

The superscripts denote actual restriction images, supported on the displayed complements. Consequently, whenever $P_0 \le P_1$ and $Q_0 \le Q_1$,

$$H(P_0, Q_1) \cap H(P_1, Q_0) = H(P_0, Q_0). \tag{3}$$

Indeed an intersection element preserves the intersection $X_0 \cup Y_0$ of the two large symmetric supports. Its complementary actions belong to the images of P_0 and Q_0, respectively. These three sets partition the domain, so (2) proves its membership in the right-hand side. The reverse inclusion is immediate.

Every interval ending at q is a point renaming of an old interval ending at $(a\ c)$: rename its point c as b. Every interval beginning at q comes similarly from the old interval beginning at $(c\ b)$. These and intervals not containing q are C. Induct on the length of an interval containing q and marks on both sides. Its end deletions are C by induction. Their groups are the two groups on the left of (3), and their common middle is the right-hand side. The acquired standard end criterion proves that interval C, and finally proves the whole marking C.

For the full transitive L, T, formula (2) is $\mathrm{Sym}(A \cup B)$. Its omission at q is the supported product of those two groups, so q remains a pure perfect cut. Subdivision of q recovers the original marking literally after naming the new point c. □

Lemma 6 (Transport of pure cuts)**.** *Subdivision of a specified pure perfect cut at i replaces its pure-cut index by $i, i+1$. All other pure perfect cuts are retained, with the later indices shifted by one, and there are no further pure perfect cuts. Contraction in Lemma 5 reverses this bookkeeping.*

Proof. Write the old cut $q = (a\ b)$ and its new marks $u = (a\ c)$, $v = (c\ b) = uqu$, with c new. For an old index $j \neq i$, the new maximal omission is literally $\langle G_j, u\rangle$, because $q = uvu$. Its orbits are the old orbits, with c adjoined to the one containing a: adjoining a transposition at a new point joins exactly that point and that one old orbit.

For $j < i$, an old perfect cut has a in its right orbit, since its suffix contains q and is supported there. The new prefix is unchanged and its suffix is the old suffix enlarged by u. Thus the cut remains perfect, with the same crossing mark and internal factors. Conversely a new perfect cut before i has a, c in its right orbit, since its suffix contains u. The orbit identity shows that deleting c gives precisely the old supported transitive suffix. The old prefix and crossing mark are unchanged. This proves the converse as well, including purity. Reverse the argument for $j > i$. The two inserted cuts are pure perfect cuts directly from their supported wings. Contraction is the literal inverse construction of Lemma 5, so has the reverse index rule. □

Theorem 7 (Stable counting)**.** *Let $\Sigma^{\kappa}(n)$ be the rank-$n-\kappa$ string C-markings of S_n, modulo marked-group isomorphism and reversal of all labels. For every $\kappa \ge 1$ its cardinality is constant for $n \ge 2\kappa + 3$.*

Proof. First use ordered point-conjugacy classes and $n \ge 7$. Put $r = n - \kappa$. The domain inequality is $2r \ge n+3$. Theorem 2 supplies an internal pure perfect cut. Define E by subdividing the least such index. The acquired pure-split theorem proves the new marking C. It generates

S_{n+1} because it contains the old S_n fixing the new point and a transposition joining that point to the old domain. Its rank is $r+1$.

Conversely a target of degree $N = n+1$ and rank $R = r+1$ satisfies $2R \geq N+4$. Theorem 2 and Lemma 4 supply two internal pure perfect cuts starting at its least internal pure index. Define D by their contraction. Lemma 5 proves actual IP and full symmetric generation. The contracted parameters satisfy $2(R-1) \geq (N-1)+3$, the exact domain condition. Its retained cut stays internal because both original cuts were internal. Lemma 6 preserves the least internal pure index under either operation. Consequently, as ordered markings up to renaming the inserted point,

$$D(E(X)) = X, \qquad E(D(Y)) = Y. \tag{4}$$

We must also pay reversal. By Lemma 3, all pure perfect cuts belong to one consecutive path. Subdividing any internal edge of this path gives the same longer path with the same endpoints, in its final ordered labels. All marks outside the interval have the same positions and actions. The explicit conjugacy fixes every point outside the internal path vertices, and matches the vertices in order from the same first endpoint. It intertwines each marked permutation. Hence extension at any two internal pure cuts gives the same ordered isomorphism class. Reversal reflects the interval and sends subdivision at i to subdivision at $r-1-i$ in the reversed old marking. The independence of the pure choice therefore makes E commute with reversal on ordered classes. Its inverse D does too. Equations (4) descend to the quotient by reversal, including self-dual classes.

For completeness, ordered marked-group isomorphism is point conjugacy for S_n when $n = 5$ or $n \geq 7$. Among its nonidentity involution classes the transpositions have uniquely smallest size. For a product of $k \geq 2$ disjoint transpositions, the ratio of its class size to the transposition class size is

$$\frac{(n-2)!}{2^{k-1}k!(n-2k)!} > 1.$$

For $k = 2$ this follows immediately for $n \geq 5$; for $k = 3$ use $n \geq 7$; for $k \geq 4$ the minimum occurs at $n = 2k$, starts at $6!/(8 \cdot 4!) > 1$, and increases with k. Thus every automorphism preserves transpositions. The maximal pairwise noncommuting families of at least four transpositions are exactly the stars at one point: a pairwise intersecting family of edges is a star or a triangle. Automorphisms therefore permute the point stars, and their pairwise intersections identify each transposition. Since these generate S_n, the automorphism is conjugation by that point permutation.

The only transitions involving degree five or six in the stated range have $\kappa = 1$. Their acquired complete extremal classification is the simplex up to marked-group isomorphism. Choose its ordinary action; the outer degree-six image is the same marked-group class. Subdivision and contraction give the successive simplex classes, so (4) and reversal hold there as well. This pays all isomorphism conventions.

We have proved $|\Sigma^\kappa(n)| = |\Sigma^\kappa(n+1)|$ at every $n \geq 2\kappa+3$. Iteration gives the assertion. □

Attribution and scope. The counting theorem and the rank and degree operation are due to Cameron, Fernandes and Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2. The acquired classifications and finite boundaries retain their original attributions. The present argument uses the repaired leaf criterion, the completed sufficient-split proof, backward propagation of its full kernels, and the interval of pure cuts. It pays both inverse equations and the quotient by reversal without asserting that the printed unrestricted extension criterion is valid. The standard end criterion and supported-transposition factorization are acquired with their earlier proofs. No numerical stable constants are computed here; their exact base classifications remain separate. This manuscript is a proof component pending distinct review, with no whole-original, new-conjecture, Lean or compression credit assigned.